DTC P0401:00 [SKYACTIV-D 1.5]

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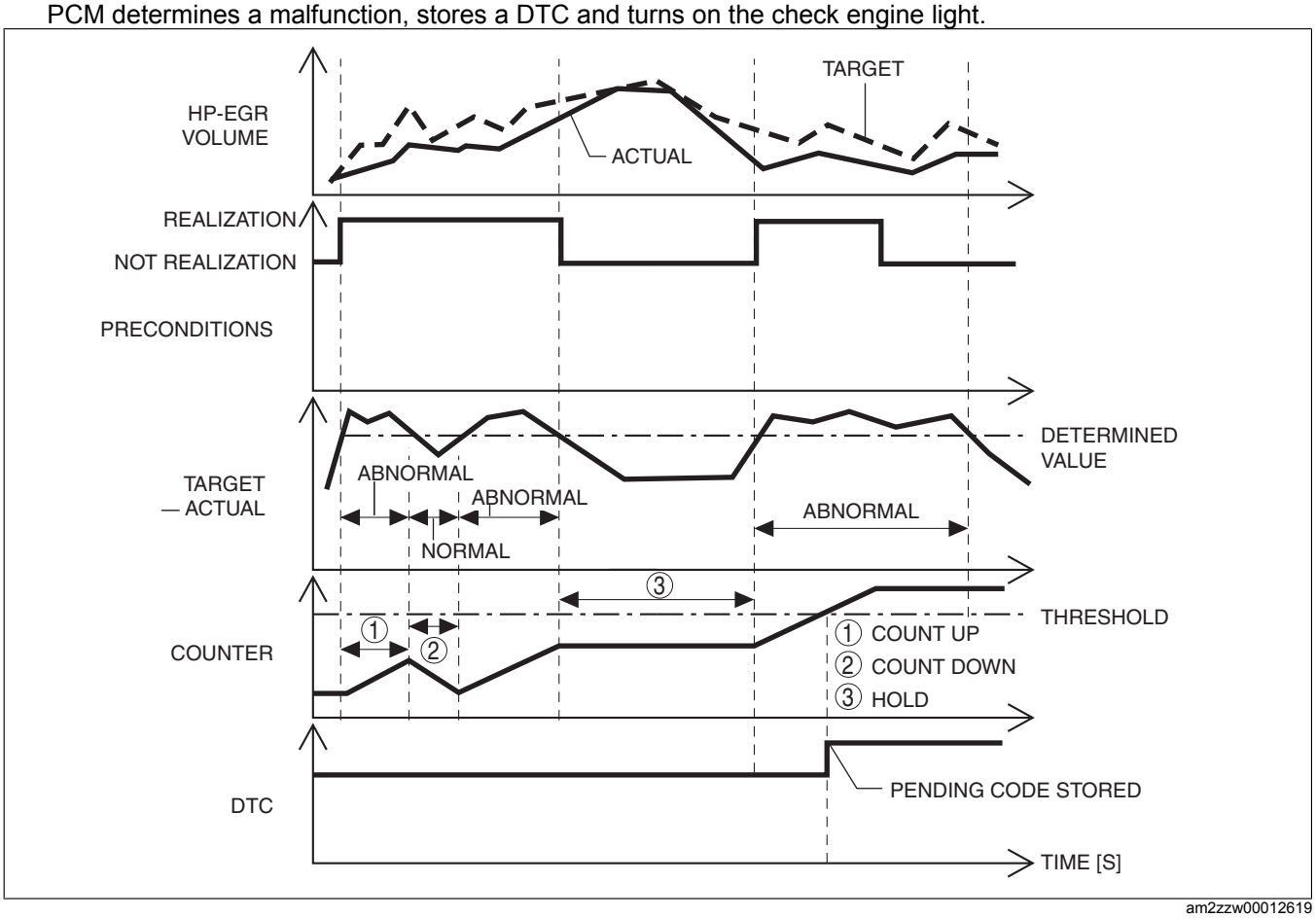
Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

Details On DTCs

DESCRIPTION	HP-EGR flow insufficient detected	
DETECTION CONDITION	Determination conditions	• When the following condition is met, the HP-EGR volume is lower than the specification for the target value for a continuous specified time.
	Preconditions	• During HP-EGR control • Engine speed: **1,200—2,500 rpm** • Fuel injection amount:**4—50 mm^3/stoke** • Engine torque: **20—240 N·m** • The following DTCs are not detected: — MAP sensor: P0107:00, P0108:00 — MAF sensor: P0102:00, P0103:00 — Exhaust gas pressure sensor No.1: P0472:00, P0473:00 — Exhaust gas temperature sensor No.1: P0546:00 — Intake shutter valve control: P2101:00, P2118:00, P1589:00 — Intake shutter valve position sensor: P0122:00, P0123:00 — HP-EGR control valve DC motor control system: P0404:00, P0488:00 — HP-EGR control valve position sensor: P0405:00, P0406:00 — LP-EGR EGR control valve DC motor control system: P045B:00, P213B:00 — LP-EGR control valve position sensor: P0407:00, P0408:00 — Exhaust shutter valve DC motor control system: P0475:00, P0476:00 — Exhaust shutter valve position sensor: P048D:00, P048E:00 — Internal PCM malfunction: P062B:00 — Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
	Drive cycle	• **2**
	Self test type	• CMDTC self test
	Sensor used	• HP-EGR control valve • Intake shutter valve • LP-EGR control valve • Exhaust shutter valve • MAF sensor • MAP sensor • Boost air temperature sensor • Exhaust gas pressure sensor No.1 • Exhaust gas temperature sensor No.1 etc.
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function.	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Check engine light is illuminated	
POSSIBLE CAUSE	• Erratic signal to PCM — MAF sensor signal malfunction — MAP sensor signal malfunction — Boost air temperature sensor signal malfunction — Exhaust gas pressure sensor No.1 signal malfunction — Exhaust gas temperature sensor No.1 signal malfunction — Input signal part connector or terminals malfunction — Input signal part related circuit malfunction • Intake shutter valve malfunction • HP-EGR control valve malfunction (stuck close) • Air suction in intake air system between turbocharger and intake manifold • HP-EGR system passage malfunction (restriction) • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM performs diagnosis of the HP-EGR system by comparing the target EGR volume with the actual.
- The PCM monitors the flow volume during HP-EGR implementation.
- The PCM calculates the difference between the target EGR flow volume and the actual EGR flow volume during HP-EGR implementation. If the monitored, actual flow volume is lower than the specification for the target flow volume, the PCM determines that there is inadequate flow volume.
- The PCM performs diagnosis when each of the preconditions during drive cycle is met. If a malfunction is detected, the malfunction detection counter begins counting up. The value of the malfunction detection counter is maintained while the preconditions are not being met. If the cumulative value of the malfunction detection counter exceeds the threshold during the first drive cycle, a malfunction is determined and a pending code is stored.
- If the calculated value of the malfunction detection counter during one drive cycle exceeds the threshold, the PCM determines a malfunction, stores a DTC and turns on the check engine light.



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Repeatability Verification Procedure

Caution
- **When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.**
- **While performing this step, always operate the vehicle in a safe and lawful manner.**

1. Start the engine at the engine cold condition.
2. Confirm that the low engine coolant temperature indicator light (blue) illuminates.
3. Drive the vehicle for 1 min at a speed of 100 km/h {62.2 mph} *1, 50 km/h {32 mph} *2 or more.
4. Repeat Step 3 operation above 2 times.

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
EXHPRES1	Exhaust gas pressure (No.1)	kPa, Bar, psi	• Idle (after warm up): Approx. 100 kPa {1 bar, 14.5 psi} • Racing (engine speed above 2,000 rpm): Approx. 125 kPa {1.27 kgf/cm^2, 18.1 psi} • Racing (engine speed above 4,000 rpm): Approx. 245 kPa {2.50 kgf/cm^2, 35.5 psi}
EXHTEMP1	Exhaust gas temperature (No.1)	°C, °F	Displays the exhaust gas temperature
CACT11	Boost air temperature	°C, °F	Displays the boost air temperature
MAF	Mass air flow	g/sec	• Switch ignition ON (engine off): Approx. 0.4 g/sec {0.13 lb/min} • Idle (after warm up): Approx. 3.2 g/sec {0.42 lb/min } • Racing (engine speed above 2,000 rpm): Approx. 9.2 g/sec {1.12 lb/min} • Racing (engine speed above 4,000 rpm): Approx. 70 g/sec {9.3 lb/min}
MAP	Manifold absolute pressure	kPa, Bar, psi	• Switch ignition ON (engine off): Approx. 100 kPa {1 bar, 14.5 psi} • Idle: Approx. 80 kPa {0.8 bar, 12 psi}

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P0401:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY OTHER RELATED DTCs OCCURRING** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the other PENDING CODE/DTC also present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**PURPOSE: VERIFY IF THERE IS PID ITEM CAUSING DRASTIC CHANGES OF ACCELERATION FLUCTUATION BY INPUT SIGNAL TO PCM** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EXHPRES1 — EXHTEMP1 — CACT11 — MAF — MAP • Is there any signal that is far out of specification?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.
6	**PURPOSE: VERIFY CONNECTOR CONNECTIONS** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **PCM:** — EXHPRES1 — EXHTEMP1 — CACT11 — MAF — MAP • When the following parts are shaken, does the PID value include a PID item which has changed? — Exhaust gas pressure sensor No.1 — Exhaust gas temperature sensor No.1 — Charge air temperature sensor — MAF sensor — MAP sensor — PCM	Yes	Repair or replace the applicable connector parts. Go to the troubleshooting procedure to perform the procedure from Step 5.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1—2
  - — Perform inspection of each separate part.
- Step 3—4
  - — Perform inspection of HP-EGR control valve passage
- Step 5—6
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT INTAKE SHUTTER VALVE CONTROL SYSTEM OPERATION** • Perform the Intake Shutter Valve Operation Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 5.
		No	Go to the next step.
2	**PURPOSE: INSPECT HP-EGR VALVE CONTROL VALVE CONTROL SYSTEM OPERATION** • Perform the HP-EGR Control Valve Operation Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 5.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**PURPOSE: INSPECT INTAKE AIR SYSTEM FOR AIR SUCTION** • Inspect for air leakage at the following: — Around connection of turbocharger and intake manifold **Note** • Engine speed may change when rust penetrating agent is sprayed on the air suction area. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 5.
		No	Go to the next step.
4	**PURPOSE: INSPECT FOR RESTRICTION OR CLOGGED IN HP-EGR PASSAGE** • Switch the ignition off. • Remove the HP-EGR valve. (See HP-EGR CONTROL VALVE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].) • Visually inspect the HP-EGR passage for clogging and the gasket correctly installed. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to the next step. (If there is clogging caused by soot in the high pressure EGR valve, inspect around the high pressure EGR piping and clean or replace it.)
		No	Go to the next step.
5	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
6	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.